AN EXPLICIT FIXED-PARAMETER OBSTRUCTION TO SUBTREE SHUFFLING

ANONYMOUS FOR REVIEW

ABSTRACT. Bisi, Dyszewski, Gantert, Johnston, Prochno, and Schmid conjectured that, for fixed integers $d \geq 2$ and $p \geq 1$, the constant function on the set of sufficiently large planar d-ary Catalan trees lies in the span of the indicators of length-p subtree-shuffle classes.

We give an explicit fixed-parameter obstruction. For $d = 7$, $p = 15$, and every $m \geq 1$, the constant function on the 7-Catalan trees with $3m$ internal vertices does not lie in the span of the length-15 shuffle indicators. If $M_{7,15,3m}$ is the tree-by-shuffle-class incidence matrix, we construct an integer vector w_m satisfying

$$M_{7,15,3m}^{\mathsf{T}} w_m = 0, \qquad \mathbf{1}^{\mathsf{T}} w_m = (7!)^{3m}(-1)^m 2^m \binom{3m+1}{m} \neq 0.$$

The construction starts from a three-dimensional noninjective Keller map. A weighted Horner suspension produces a noninjective map $\Phi = I - H : \mathbb{C}^{15} \to \mathbb{C}^{15}$ for which H is homogeneous of degree seven and $(JH)^{15} = 0$. An inverse-slice identity and Lagrange inversion give infinitely many nonzero coefficients of the formal inverse of Φ. The shuffle lemma and tree-inverse formula then turn those coefficients into the separating vectors above.

The same fixed pair disproves the corresponding shuffle-chain conjecture and the all-degree extension of Singer's full-span conjecture. The separator vanishes on every 15-perfect tree, so the exact obstruction is supported on the exponentially rare exceptional family; this is compatible with the previously proved asymptotic approximation theorem.

1. INTRODUCTION

Let $\mathcal{C}_k^{(d)}$ be the set of rooted planar full d-ary trees with k internal vertices. Given an ancestral path of length p, one may rearrange the $(d-1)p$ rooted subtrees attached to the siblings along that path. The set of distinct trees obtainable in this way is a length-p shuffle class. Bisi et al. introduced the following conjecture [2, Conjecture 1.3].

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Subtree-shuffling conjecture. For every $d \geq 2$ and $p \geq 1$, there is $k_0(d,p)$ such that, for every $k \geq k_0(d,p)$,

$$\mathbf{1} \in \operatorname{span}\{\mathbf{1}_S : S \text{ is a length-}p \text{ shuffle class in } \mathcal{C}_k^{(d)}\}.$$

The conjecture was designed as an unlabelled-tree route toward the Jacobian conjecture. Its origin lies in Singer's Catalan-tree approach [6, 7]. The subtree-shuffling preprint was posted on 19 January 2023 and appeared in final journal form on 22 January 2026. The fixed certificate below is dated 23 July 2026, 1,281 days after the preprint and 182 days after journal publication.

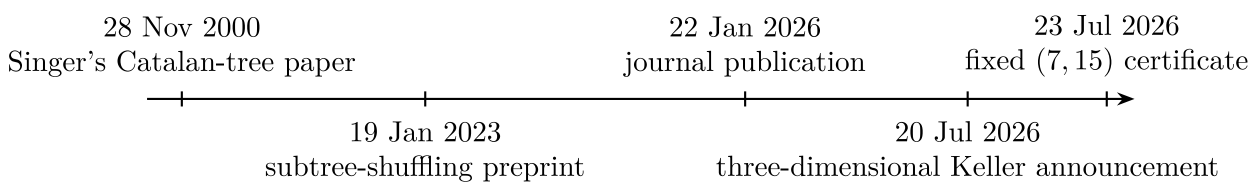

FIGURE 1. The combinatorial line of development. The present result does not decide Singer's original binary-tree conjecture; it refutes the later all-degree extension at $d = 7$.

Our main result is the following fixed-parameter negation.

Theorem 1.1. *For every integer $m \geq 1$,*

$$\mathbf{1} \notin \operatorname{span}\{\mathbf{1}_S : S \text{ is a length-}15 \text{ shuffle class in } \mathcal{C}_{3m}^{(7)}\}.$$

Consequently, the subtree-shuffling conjecture is false for the fixed pair $(d,p) = (7,15)$.

The proof supplies a dual certificate. Let $M_{d,p,k}$ denote the matrix whose rows are indexed by $T \in \mathcal{C}_k^{(d)}$, whose columns are indexed by the distinct length-p shuffle classes, and whose entries are

$$(M_{d,p,k})_{T,S} = \mathbf{1}_{\{T \in S\}}.$$

Theorem 1.2. *For every $m \geq 1$, there exists an integer vector $w_m \in \mathbb{Z}^{\mathcal{C}_{3m}^{(7)}}$ such that*

$$M_{7,15,3m}^{\mathsf{T}} w_m = 0$$

and

$$\mathbf{1}^{\mathsf{T}} w_m = (7!)^{3m}(-1)^m 2^m \binom{3m+1}{m} \neq 0.$$

The crucial point is explicitness. A noninjective Keller map makes the universal subtree-shuffling conjecture logically untenable once combined with the implication of Bisi et al.; the present argument isolates a single fixed degree, a single fixed path length, infinitely many tree sizes, and an exact left-kernel vector.

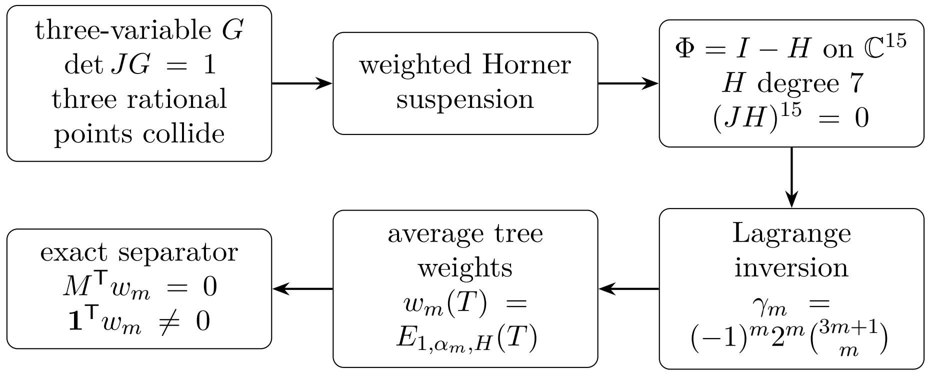

FIGURE 2. The proof pipeline.

2. Shuffle duality and the separation principle

We recall the two ingredients from [2] that translate formal inversion into unlabelled-tree linear algebra.

Let $H = (H_1, \dots, H_n)$ be homogeneous of degree d, written in factorial normalization as

$$H_i(X) = \sum_{|\beta|=d} H_{i,\beta} \frac{X^\beta}{\beta!}.$$

For $T \in \mathcal{C}_k^{(d)}$, a root type $i \in [n]$, and a leaf-type multi-index $\alpha \in \mathbb{Z}_{\geq 0}^n$ of size $(d-1)k+1$, let $E_{i,\alpha,H}(T)$ be the sum of the H-weights of all type-labellings of T having root type i and leaf multiplicities α.

The shuffle lemma [2, Lemma 3.3] states that if $(JH)^p = 0$, with $1 \leq p \leq n$, then

$$\sum_{T \in S} E_{i,\alpha,H}(T) = 0 \tag{1}$$

for every length-p shuffle class S.

If $F = I - H$, then its formal inverse has an expansion

$$(F^{-1})_i(X) = X_i + \sum_{k \geq 1} \sum_{|\alpha|=(d-1)k+1} g_{i,\alpha} X^\alpha,$$

and the tree-inverse formula is

$$g_{i,\alpha} = \frac{1}{(d!)^k} \sum_{T \in \mathcal{C}_k^{(d)}} E_{i,\alpha,H}(T). \tag{2}$$

Proposition 2.1 (Separation principle)**.** *Suppose H is homogeneous of degree d, $(JH)^p = 0$, and $g_{i,\alpha} \neq 0$ for some $|\alpha| = (d-1)k+1$. Define*

$$w(T) = E_{i,\alpha,H}(T), \qquad T \in \mathcal{C}_k^{(d)}.$$

Then

$$M_{d,p,k}^{\mathsf{T}} w = 0, \qquad \mathbf{1}^{\mathsf{T}} w = (d!)^k g_{i,\alpha} \neq 0.$$

In particular, $\mathbf{1} \notin \operatorname{col} M_{d,p,k}$.

Proof. Equation (1) is precisely $M_{d,p,k}^{\mathsf{T}} w = 0$. Equation (2) gives the total sum. If $\mathbf{1} = M\lambda$, then

$$\mathbf{1}^{\mathsf{T}} w = \lambda^{\mathsf{T}} M^{\mathsf{T}} w = 0,$$

a contradiction. □

3. A normalized three-dimensional Keller map

Consider $G = (A, B, C) : \mathbb{C}^3 \to \mathbb{C}^3$ given by

$$A = x - 3x^2y - x^3z, \tag{3}$$

$$B = y + 3xz + 24xy^2 + 12x^2yz + 36x^2y^3 + 12x^3y^2z, \tag{4}$$

$$C = z + 8y^2 + 6xyz + 28xy^3 + 12x^2y^2z + 24x^2y^4 + 8x^3y^3z. \tag{5}$$

This is the identity-linear determinant-one normalization of the three-dimensional map discussed in 2026; a contemporaneous account of the announcement appears in [5]. The exact source and target normalization is recorded in the reproducibility files.

Proposition 3.1. *The map G satisfies* $\det JG = 1$*, and*

$$G\left(0, 0, -\frac{1}{8}\right) = G\left(1, -\frac{3}{4}, \frac{13}{4}\right) = G\left(-1, \frac{3}{4}, \frac{13}{4}\right) = \left(0, 0, -\frac{1}{8}\right).$$

Proof. The collisions follow by direct substitution. For the determinant, put

$$u = 1 + 2xy, \qquad r = \frac{x}{u}.$$

On the dense open set $u \neq 0$, direct simplification gives

$$B = y + 3rC, \qquad A = r - r^2B + 2r^3C. \tag{6}$$

The coordinate change $(x, y, z) \mapsto (r, y, C)$ has determinant

$$\frac{\partial r}{\partial x}\frac{\partial C}{\partial z} = \frac{1}{u^2}u^3 = u.$$

The change $(r, y, C) \mapsto (C, B, r)$ has determinant -1. Holding B, C fixed in the second identity of (6) gives

$$\frac{\partial A}{\partial r} = 1 - 2rB + 6r^2C = 1 - 2ry = \frac{1}{u}.$$

Thus the determinant of the map to (C, B, A) is -1. Swapping the first and third target coordinates gives $\det JG = 1$ on $u \neq 0$, hence everywhere by polynomial continuation. □

4. Weighted Horner suspension

We next give the construction that turns a nonhomogeneous identity-linear Keller map into a homogeneous Keller map while retaining a controlled nilpotency index.

Let

$$G_i(X) = X_i + \sum_{j=2}^{d_i} K_{i,j}(X),$$

where $K_{i,j}$ is homogeneous of degree j. Introduce auxiliary variables $Y_{i,3}, \ldots, Y_{i,d_i}$ and let

$$M = n + \sum_{i=1}^{n} (d_i - 2).$$

Define $N : \mathbb{C}^M \to \mathbb{C}^M$ by

$$N_{X_i} = K_{i,2}(X) + Y_{i,3}, \tag{7}$$

$$N_{Y_{i,j}} = -K_{i,j}(X) - Y_{i,j+1}, \qquad 3 \le j < d_i, \tag{8}$$

$$N_{Y_{i,d_i}} = -K_{i,d_i}(X). \tag{9}$$

Theorem 4.1 (Weighted Horner suspension)**.** *Assume* $\det JG = 1$*. Then*

$$(JN)^M = 0.$$

Let $d = \max_i d_i$*, add one variable* h*, and define*

$$\mathcal{H}(Z, h) = \left(h^d N \left(\frac{Z}{h} \right), 0 \right).$$

Then $\mathcal{H}$ *is homogeneous of degree* d *and*

$$(J\mathcal{H})^{M+1} = 0.$$

Consequently, $\Phi = I + \mathcal{H}$ *has Jacobian determinant one.*

Moreover, every collision of G *lifts to a collision of* Φ *on the hyperplane* $h = 1$*.*

Proof. For an indeterminate s, put $U_s = I + sN$. Thus

$$X_i' = X_i + sK_{i,2}(X) + sY_{i,3},$$
$$Y_{i,j}' = Y_{i,j} - sK_{i,j}(X) - sY_{i,j+1},$$
$$Y_{i,d_i}' = Y_{i,d_i} - sK_{i,d_i}(X).$$

The weighted Horner identity is

$$X_i' - \sum_{j=3}^{d_i} s^{j-2} Y_{i,j}' = X_i + \sum_{j=2}^{d_i} s^{j-1} K_{i,j}(X) = s^{-1} G_i(sX). \tag{10}$$

The operation on the left is a target shear of determinant one. After this shear, the original-coordinate outputs depend only on X, while the auxiliary-output Jacobian with respect to the auxiliary variables is unit triangular. Hence

$$\det(I + sJN) = \det J\big(s^{-1} G(sX)\big) = \det JG(sX) = 1.$$

It follows that the characteristic polynomial of JN is λ^M. Cayley–Hamilton gives $(JN)^M = 0$.

The expression $h^d N(Z/h)$ is homogeneous of degree d. On $h \neq 0$, its Jacobian has block form

$$J\mathcal{H} = \begin{pmatrix} h^{d-1} JN(Z/h) & * \\ 0 & 0 \end{pmatrix}.$$

The $(M+1)$st power vanishes because $(JN)^M = 0$. Since the entries are polynomials, the identity extends to $h = 0$.

For the collision statement, at $h = 1$ set

$$Y_{i,j} = \sum_{\ell=j}^{d_i} K_{i,\ell}(q).$$

Then all auxiliary outputs telescope to zero, and the ith original output is $G_i(q)$. Equal G-images therefore yield equal Φ-images. $\square$

The construction also controls a slice of the formal inverse.

Proposition 4.2 (Inverse slice)**.** *Let π_X project onto the original n coordinates. If all auxiliary target coordinates vanish, then*

$$\pi_X \Phi^{-1}(A, 0, h) = h^{-(d-2)} G^{-1}\big(h^{d-2} A\big)$$

as an identity of formal power series.

Proof. Write $Z = hz$ and $s = h^{d-1}$. Then

$$\Phi(Z, h) = \big(hU_s(z), h\big).$$

When the auxiliary target coordinates are zero, the target shear in (10) leaves the original target coordinates unchanged. Thus

$$s^{-1} G(sx) = \frac{A}{h}, \qquad G(sx) = h^{d-2} A.$$

Therefore $sx = G^{-1}(h^{d-2}A)$, while the original coordinates of Z are hx. Since $h/s = h^{-(d-2)}$, the formula follows. $\square$

5. The explicit homogeneous map in fifteen variables

For (3)–(5), the component degrees are

$$(d_1, d_2, d_3) = (4, 6, 7),$$

so $M = 14$ before homogenization and $M + 1 = 15$ afterwards. Use coordinates

$$x, y, z, a_3, a_4, b_3, b_4, b_5, b_6, c_3, c_4, c_5, c_6, c_7, h.$$

The suspended map is

$$\begin{aligned}
\Phi_x &= x + a_3 h^6, \\
\Phi_y &= y + 3xzh^5 + b_3 h^6, \\
\Phi_z &= z + 8y^2h^5 + c_3 h^6, \\
\Phi_{a_3} &= a_3 + 3x^2yh^4 - a_4 h^6, \\
\Phi_{a_4} &= a_4 + x^3zh^3, \\
\Phi_{b_3} &= b_3 - 24xy^2h^4 - b_4 h^6, \\
\Phi_{b_4} &= b_4 - 12x^2yzh^3 - b_5 h^6, \\
\Phi_{b_5} &= b_5 - 36x^2y^3h^2 - b_6 h^6, \\
\Phi_{b_6} &= b_6 - 12x^3y^2zh, \\
\Phi_{c_3} &= c_3 - 6xyzh^4 - c_4 h^6, \\
\Phi_{c_4} &= c_4 - 28xy^3h^3 - c_5 h^6, \\
\Phi_{c_5} &= c_5 - 12x^2y^2zh^2 - c_6 h^6, \\
\Phi_{c_6} &= c_6 - 24x^2y^4h - c_7 h^6, \\
\Phi_{c_7} &= c_7 - 8x^3y^3z, \\
\Phi_h &= h.
\end{aligned}$$

Corollary 5.1. *Write $\Phi = I - H$. Then H is homogeneous of degree seven,*

$$(JH)^{15} = 0, \qquad \det J\Phi = 1,$$

and Φ is not injective.

Proof. Homogeneity and nilpotency follow from Theorem 4.1. Noninjectivity follows by lifting the three points in Proposition 3.1. For completeness, put

$$Q_0 = \left(0, 0, -\frac{1}{8}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1\right)$$

and, for $\varepsilon \in \{1, -1\}$,

$$\begin{aligned}
Q_\varepsilon = \Big(&\varepsilon, -\frac{3\varepsilon}{4}, \frac{13}{4}, -\varepsilon, -\frac{13\varepsilon}{4}, -9\varepsilon, -\frac{45\varepsilon}{2}, \frac{27\varepsilon}{4}, \frac{351\varepsilon}{16}, \\
&-\frac{63}{8}, \frac{27}{4}, \frac{297}{16}, -\frac{27}{8}, -\frac{351}{32}, 1\Big).
\end{aligned}$$

Direct substitution gives

$$\Phi(Q_0) = \Phi(Q_1) = \Phi(Q_{-1}) = Q_0.$$

□

6. A nonzero inverse coefficient

Let $(A,B,C)=G(x,y,z)$ and put $r=x/(1+2xy)$. On the target plane $B=0$, (6) becomes

$$A=r+2Cr^3, \qquad y=-3Cr. \tag{11}$$

Moreover,

$$x=\frac{r}{1+6Cr^2}=r\frac{\partial r}{\partial A}=\frac{1}{2}\frac{\partial(r^2)}{\partial A}.$$

Proposition 6.1. *For every $m\geq 0$,*

$$[A^{2m+1}C^m](G^{-1})_1=\gamma_m:=(-1)^m2^m\binom{3m+1}{m}.$$

In particular, $\gamma_m\neq 0$.

Proof. Equation (11) is equivalent to

$$r=A(1+2Cr^2)^{-1}.$$

The Lagrange–Bürmann formula gives

$$[A^{2m+2}]r^2=\frac{2}{2m+2}[t^{2m}](1+2Ct^2)^{-(2m+2)}=\frac{(-1)^m2^m}{m+1}\binom{3m+1}{m}C^m.$$

Applying $\frac{1}{2}\partial/\partial A$ gives the result. □

By Proposition 4.2, with $d=7$,

$$(\Phi^{-1})_1(A,0,h)=h^{-5}(G^{-1})_1(h^5A).$$

Consequently,

$$[A^{2m+1}C^mh^{15m}](\Phi^{-1})_1=\gamma_m. \tag{12}$$

The total degree is

$$(2m+1)+m+15m=18m+1=6(3m)+1,$$

which is the leaf count of a 7-ary Catalan tree with $3m$ internal vertices.

7. Proof of the fixed-parameter obstruction

Set

$$\alpha_m=(2m+1,0,m,\underbrace{0,\dots,0}_{\text{11 auxiliary types}},15m)\in\mathbb{Z}^{15}_{\geq 0}.$$

Then $|\alpha_m|=18m+1$. For $T\in\mathcal{C}^{(7)}_{3m}$, define

$$w_m(T)=E_{1,\alpha_m,H}(T), \tag{13}$$

where $\Phi=I-H$ is the explicit map of the previous section.

Proof of Theorem 1.2. By Corollary 5.1, $(JH)^{15} = 0$. The shuffle lemma with $n = p = 15$ gives

$$\sum_{T \in S} w_m(T) = 0$$

for every length-15 shuffle class S, hence

$$M_{7,15,3m}^{\mathsf{T}} w_m = 0.$$

By the tree-inverse formula and (12),

$$\begin{aligned} \mathbf{1}^{\mathsf{T}} w_m &= \sum_{T \in \mathcal{C}_{3m}^{(7)}} E_{1,\alpha_m,H}(T) \\ &= (7!)^{3m} [A^{2m+1} C^m h^{15m}] (\Phi^{-1})_1 \\ &= (7!)^{3m} (-1)^m 2^m \binom{3m+1}{m}. \end{aligned}$$

This is nonzero.

The ordinary coefficients of H are integers. In factorial normalization, $H_{i,\beta}$ is the ordinary coefficient multiplied by $\beta!$, and is therefore an integer. Formula (13) is a finite sum of products of such integers, so w_m is integer-valued. □

Proof of Theorem 1.1. Apply Proposition 2.1 and Theorem 1.2. □

The result is not merely caused by small trees having no path of length 15.

Proposition 7.1 (Fully covered regime)**.** *Every 7-ary Catalan tree with*

$$k \geq 113037178809$$

has a path of length 15. The first such k is divisible by three, and Theorem 1.1 applies there.

Proof. If a full 7-ary tree has no path of length 15, its internal vertices can occur only at depths $0, \dots, 13$. Hence it has at most

$$1 + 7 + \cdots + 7^{13} = \frac{7^{14} - 1}{6} = 113037178808$$

internal vertices. The next integer is

$$113037178809 = 3 \cdot 37679059603.$$

□

8. Exact failure and asymptotic approximation

A 15-perfect tree contains a length-15 path whose off-path sibling subtrees are all leaves. Such a path defines a singleton shuffle class. Applying the shuffle identity to that singleton gives

$$w_m(T) = 0$$

for every 15-perfect tree. Thus

$$\operatorname{supp} w_m \subseteq \{T \in \mathcal{C}_{3m}^{(7)} : T \text{ is not 15-perfect}\}.$$

Bisi et al. prove that, for fixed d, p, the proportion of non-p-perfect trees decays exponentially in k [2, Theorem 1.7]. The exact separator is therefore supported on an exponentially rare exceptional family. This explains how exact spanning can fail while their unconditional ℓ^1 and ℓ^∞ approximation theorem remains valid.

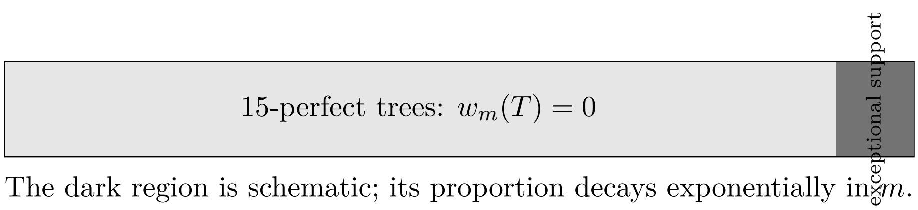

Figure 3. The exact obstruction lives only on the rare non-perfect trees.

9. Consequences and scope

Corollary 9.1. *For $d = 7$ and $p = 15$, the shuffle-chain conjecture of Bisi et al. fails for infinitely many tree sizes k, namely every $k = 3m$.*

Proof. A length-p shuffle chain with the uniform stationary distribution produces a nonnegative representation of the constant function by length-p shuffle-class indicators [2, Theorem 1.6 and its proof]. Such a representation is impossible by Theorem 1.1. □

Corollary 9.2. *The all-degree extension of Singer's full-span conjecture fails for $d = 7$, $p = 15$, and every $k = 3m$.*

Proof. Full spanning would in particular span the constant function. □

Remark 9.3. *Corollary 9.2 does not decide Singer's original binary-tree conjecture. It refutes the later formulation in which d is allowed to be arbitrary.*

The construction does not establish minimality. Natural open problems are to determine the least failing d, the least failing p for that d, the least possible homogeneous degree and dimension, and a direct unlabelled combinatorial description of the support and signs of w_m.

10. Reproducibility and verification

The accompanying repository contains:

(i) exact SymPy checks of the Jacobian determinants and collisions;
(ii) a generic implementation of the weighted Horner suspension;
(iii) symbolic checks of the telescoping identities and unit-triangular auxiliary block;
(iv) exact integer matrix-power regression checks at several specializations;
(v) fixed-point formal-series checks of the Lagrange coefficient;
(vi) tests enforcing the fixed claim $(d, p) = (7, 15)$;
(vii) a continuous-integration workflow that runs the checks and compiles this manuscript.

The matrix-power specializations are regression tests, not substitutes for the symbolic proof. The proof of nilpotency is the identity $\det(I + sJN) = 1$ together with Cayley–Hamilton and the homogenized block argument in Theorem 4.1.

Provenance and responsibility

The starting three-dimensional polynomial map was supplied in the research discussion leading to this manuscript. The normalization, weighted Horner suspension, inverse-slice identity, coefficient extraction, and fixed shuffle certificate are documented here and in the repository history. Before submission, the human contributors must settle authorship, acknowledgements, and any journal-required disclosure of AI-assisted research and drafting. The named authors bear responsibility for every mathematical and historical claim.

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